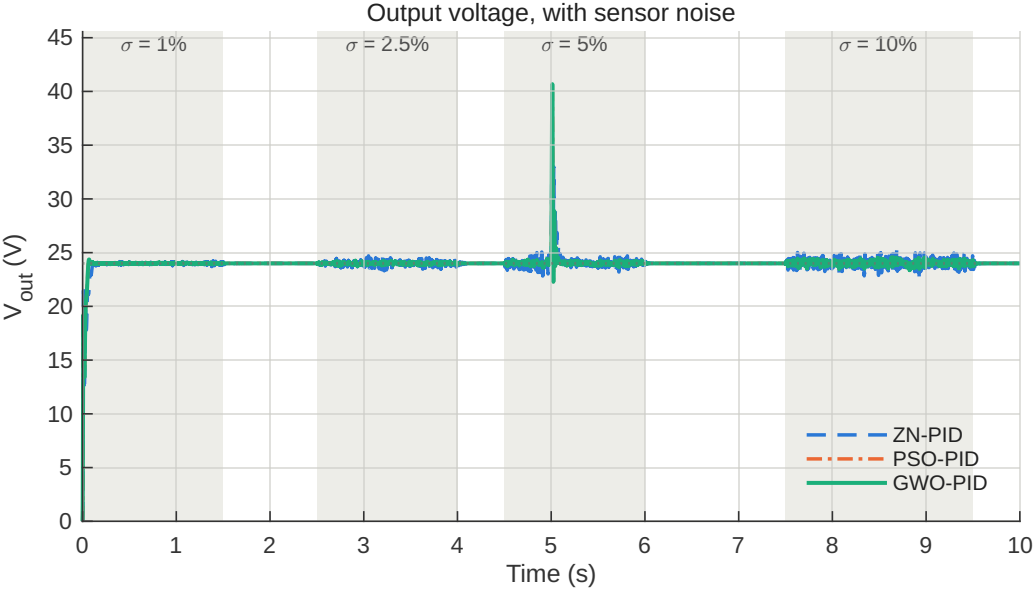

Output voltage, with sensor noise
σ = 1%
σ = 2.5%
σ = 5%
σ = 10%
ZN-PID
PSO-PID
GWO-PID
V_out (V)
Time (s)
0
5
10
15
20
25
30
35
40
45
0
1
2
3
4
5
6
7
8
9
10